



ANDROID STATIC ANALYSIS REPORT



 VulnApp (1.0)

File Name:

VulnApp.apk

Package Name:

opensecurity.vulnapp

Scan Date:

May 5, 2025, 7:11 p.m.

App Security Score:

38/100 (HIGH RISK)

Grade:



FINDINGS SEVERITY

 HIGH	 MEDIUM	 INFO	 SECURE	 HOTSPOT
4	6	0	1	1

FILE INFORMATION

File Name: VulnApp.apk

Size: 0.9MB

MD5: 5f6540793305ef3c8376669b7e150297

SHA1: 053844a5bed8eb0bf3f087328f8cd09d972f7aac

SHA256: e38038adbeff00f9766f7daeea1b84ca10e469434a084b370e89878927b4657b

APP INFORMATION

App Name: VulnApp

Package Name: opensecurity.vulnapp

Main Activity: opensecurity.vulnapp.MainActivity

Target SDK: 21

Min SDK: 16

Max SDK:

Android Version Name: 1.0

Android Version Code: 1

APP COMPONENTS

Activities: 5

Services: 0

Receivers: 1

Providers: 0

Exported Activities: 2

Exported Services: 0

Exported Receivers: 1

Exported Providers: 0

CERTIFICATE INFORMATION

Binary is signed

v1 signature: True

v2 signature: False

v3 signature: False

v4 signature: False

X.509 Subject: CN=Ajin

Signature Algorithm: rsassa_pkcs1v15

Valid From: 2015-06-28 15:12:47+00:00

Valid To: 2040-06-21 15:12:47+00:00

Issuer: CN=Ajin

Serial Number: 0x55900eef

Hash Algorithm: sha1

md5: 0b5e12aac2b630c0510aa23e17c2655c

sha1: f78fd64754e89e79a25d58ea8c091ebe99fb375c

sha256: 40df6663b240ce93e5bfb04023feb22c235a3799a051cc7b23981e6100fe381b

sha512: 888e8007e14c05f09412abf778b216c01a34212fb3c16336784494adcb33808b4b048e50c3b71491a56649f6687e3bc230cc598294a04fa44b1c078ba0c40063

Found 1 unique certificates

APPLICATION PERMISSIONS

PERMISSION	STATUS	INFO	DESCRIPTION
android.permission.SEND_SMS	dangerous	send SMS messages	Allows application to send SMS messages. Malicious applications may cost you money by sending messages without your confirmation.



APKID ANALYSIS

FILE	DETAILS	
classes.dex	FINDINGS	DETAILS
	Compiler	dx (possible dexmerge)
	Manipulator Found	dexmerge



NETWORK SECURITY

NO	SCOPE	SEVERITY	DESCRIPTION
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CERTIFICATE ANALYSIS

HIGH: 2 | WARNING: 0 | INFO: 1

TITLE	SEVERITY	DESCRIPTION
Signed Application	info	Application is signed with a code signing certificate
Application vulnerable to Janus Vulnerability	high	Application is signed with v1 signature scheme, making it vulnerable to Janus vulnerability on Android 5.0-8.0, if signed only with v1 signature scheme. Applications running on Android 5.0-7.0 signed with v1, and v2/v3 scheme is also vulnerable.
Certificate algorithm vulnerable to hash collision	high	Application is signed with SHA1withRSA. SHA1 hash algorithm is known to have collision issues.

Q MANIFEST ANALYSIS

HIGH: 2 | WARNING: 4 | INFO: 0 | SUPPRESSED: 0

NO	ISSUE	SEVERITY	DESCRIPTION
1	App can be installed on a vulnerable unpatched Android version Android 4.1-4.1.2, [minSdk=16]	high	This application can be installed on an older version of android that has multiple unfixed vulnerabilities. These devices won't receive reasonable security updates from Google. Support an Android version => 10, API 29 to receive reasonable security updates.
2	Application Data can be Backed up [android:allowBackup=true]	warning	This flag allows anyone to backup your application data via adb. It allows users who have enabled USB debugging to copy application data off of the device.
3	Activity (opensecurity.vulnapp.ExportedActivity) is vulnerable to StrandHogg 2.0	high	Activity is found to be vulnerable to StrandHogg 2.0 task hijacking vulnerability. When vulnerable, it is possible for other applications to place a malicious activity on top of the activity stack of the vulnerable application. This makes the application an easy target for phishing attacks. The vulnerability can be remediated by setting the launch mode attribute to "singleInstance" and by setting an empty taskAffinity (taskAffinity=""). You can also update the target SDK version (21) of the app to 29 or higher to fix this issue at platform level.

NO	ISSUE	SEVERITY	DESCRIPTION
4	Activity (opensecurity.vulnapp.ExportedActivity) is not Protected. [android:exported=true]	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device.
5	Activity (opensecurity.vulnapp.ImplicitlyExportedActivity) is not Protected. An intent-filter exists.	warning	An Activity is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. The presence of intent-filter indicates that the Activity is explicitly exported.
6	Broadcast Receiver (opensecurity.vulnapp.MyReceiver) is not Protected. An intent-filter exists.	warning	A Broadcast Receiver is found to be shared with other apps on the device therefore leaving it accessible to any other application on the device. The presence of intent-filter indicates that the Broadcast Receiver is explicitly exported.

</> CODE ANALYSIS

HIGH: 0 | WARNING: 2 | INFO: 0 | SECURE: 0 | SUPPRESSED: 0

NO	ISSUE	SEVERITY	STANDARDS	FILES
1	App uses SQLite Database and execute raw SQL query. Untrusted user input in raw SQL queries can cause SQL Injection. Also sensitive information should be encrypted and written to the database.	warning	CWE: CWE-89: Improper Neutralization of Special Elements used in an SQL Command ('SQL Injection') OWASP Top 10: M7: Client Code Quality	opensecurity/vulnapp/S qliteDB.java
2	Files may contain hardcoded sensitive information like usernames, passwords, keys etc.	warning	CWE: CWE-312: Cleartext Storage of Sensitive Information OWASP Top 10: M9: Reverse Engineering OWASP MASVS: MSTG-STORAGE-14	opensecurity/vulnapp/S qliteDB.java

NIAP ANALYSIS v1.3

NO	IDENTIFIER	REQUIREMENT	FEATURE	DESCRIPTION
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BEHAVIOUR ANALYSIS

RULE ID	BEHAVIOUR	LABEL	FILES
00063	Implicit intent(view a web page, make a phone call, etc.)	control	opensecurity/vulnapp/ExportedActivity.java
00193	Send a SMS message	sms	opensecurity/vulnapp/MyReceiver.java

ABUSED PERMISSIONS

TYPE	MATCHES	PERMISSIONS
Malware Permissions	1/25	android.permission.SEND_SMS
Other Common Permissions	0/44	

Malware Permissions:

Top permissions that are widely abused by known malware.

Other Common Permissions:

Permissions that are commonly abused by known malware.

EMAILS

EMAIL	FILE
secure@gmail.com	opensecurity/vulnapp/ExportedActivity.java

SCAN LOGS

Timestamp	Event	Error
2025-05-05 19:11:40	Generating Hashes	OK
2025-05-05 19:11:40	Extracting APK	OK
2025-05-05 19:11:40	Unzipping	OK
2025-05-05 19:11:40	Parsing APK with androguard	OK
2025-05-05 19:11:40	Extracting APK features using aapt/aapt2	OK
2025-05-05 19:11:40	Getting Hardcoded Certificates/Keystores	OK

2025-05-05 19:11:42	Parsing AndroidManifest.xml	OK
2025-05-05 19:11:42	Extracting Manifest Data	OK
2025-05-05 19:11:42	Manifest Analysis Started	OK
2025-05-05 19:11:42	Performing Static Analysis on: VulnApp (opensecurity.vulnapp)	OK
2025-05-05 19:11:42	Fetching Details from Play Store: opensecurity.vulnapp	OK
2025-05-05 19:11:42	Checking for Malware Permissions	OK
2025-05-05 19:11:42	Fetching icon path	OK
2025-05-05 19:11:42	Library Binary Analysis Started	OK
2025-05-05 19:11:42	Reading Code Signing Certificate	OK
2025-05-05 19:11:42	Running APKiD 2.1.5	OK
2025-05-05 19:11:44	Updating Trackers Database....	OK

2025-05-05 19:11:44	Detecting Trackers	OK
2025-05-05 19:11:44	Decompiling APK to Java with JADX	OK
2025-05-05 19:11:52	Converting DEX to Smali	OK
2025-05-05 19:11:52	Code Analysis Started on - java_source	OK
2025-05-05 19:11:53	Android SBOM Analysis Completed	OK
2025-05-05 19:11:53	Android SAST Completed	OK
2025-05-05 19:11:53	Android API Analysis Started	OK
2025-05-05 19:11:54	Android API Analysis Completed	OK
2025-05-05 19:11:55	Android Permission Mapping Started	OK
2025-05-05 19:11:55	Android Permission Mapping Completed	OK
2025-05-05 19:11:55	Android Behaviour Analysis Started	OK

2025-05-05 19:11:56	Android Behaviour Analysis Completed	OK
2025-05-05 19:11:56	Extracting Emails and URLs from Source Code	OK
2025-05-05 19:11:56	Email and URL Extraction Completed	OK
2025-05-05 19:11:56	Extracting String data from APK	OK
2025-05-05 19:11:56	Extracting String data from Code	OK
2025-05-05 19:11:56	Extracting String values and entropies from Code	OK
2025-05-05 19:11:56	Performing Malware check on extracted domains	OK
2025-05-05 19:11:56	Saving to Database	OK

Report Generated by - MobSF v4.3.3

Mobile Security Framework (MobSF) is an automated, all-in-one mobile application (Android/iOS/Windows) pen-testing, malware analysis and security assessment framework capable of performing static and dynamic analysis.

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